Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
13	(a)			award (1) for **all** readings in the table correct Titre **A** 7.40 7.45 7.50 Titre **B** 14.90 15.00 14.85 award (1) each for means calculated mean titre **A** = 7.45 cm^3 mean titre **B** = 14.92 cm^3 accept 14.90	1	2		3	2	3
	(b)			(volumetric) pipette	1			1		1
	(c)			award (1) for any of following double the mass of solid dissolved double the amount of solution used halve the concentration of acid			1	1		1
	(d)			award (1) for any sensible suggestion colour change for titre **A** easier to see colour change for titre **B** more difficult to see only one indicator needed for titre **A** second titre depends on the first one			1	1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(e)	(i)		7.45×10^{-4} mol in $25cm^3$ 7.45×10^{-3} mol in original solution		1		1		1
		(ii)		moles of HCl in titre **B** = 1.492×10^{-3} (1) moles of $NaHCO_3$ in $25cm^3$ = $(1.492 \times 10^{-3}) - (7.45 \times 10^{-4})$ $= 7.47 \times 10^{-4}$ moles of $NaHCO_3$ in original solution = 7.47×10^{-3} (1)		2		2	1	2
		(iii)		moles of $NaHCO_3$ ≈ moles of Na_2CO_3 therefore $x = 1$			1	1		
	(f)			M_r of anhydrous $Na_2CO_3.NaHCO_3$ = 190 (1) M_r of water present in formula = 226 – 190 = 36 $\frac{36}{18} = 2$ therefore $y = 2$ (1)		2		2	1	2
				Question 13 total	**2**	**7**	**3**	**12**	**4**	**11**